

Lecture 14 (2026-08-09)

Quick Summary

- **Break-even Analysis:** Calculation of Break-even Point (BEP) in units and sales value, and determining target units to reach a specific profit.
- **Profit Volume (P/V) Ratio:** Calculation of the percentage of contribution relative to sales.
- **Margin of Safety (MoS):** Quantifying the cushion between current sales and the break-even point.
- **Multi-product Analysis:** Handling products with different contribution margins using a "bundle" or "weighted average" approach.
- **Sensitivity Analysis:** Evaluating how changes in selling price or variable costs impact the Break-even Point.
- **Operating Leverage:** Measuring the proportion of fixed costs in total costs and the resulting impact on profit volatility.
- **Indifference Point:** Determining the point where two different options (e.g., two machines) yield the same profit.
- **Make or Buy Decisions:** Evaluating whether to produce a component internally or purchase it from an external supplier based on relevant costs.

Key Definitions

- **Contribution:** The difference between selling amount and variable cost per unit (Selling amount - Variable Cost).
- **Profit Volume (P/V) Ratio:** The ratio of contribution to sales, expressed as a percentage.
- **Margin of Safety (MoS):** The amount of sales (in units or value) that exceeds the break-even point.
- **Bundle Contribution:** In multi-product scenarios, the combined contribution of a specific mix of products (e.g., a 3:2 ratio).
- **Operating Leverage:** The ratio of contribution to profit. It indicates the proportion of fixed costs in the total cost structure.
- **Indifference Point:** The point at which two different alternatives provide the same profit/loss.
- **Sunk Cost:** A cost that has already been incurred and cannot be changed by any future decision (e.g., "already incurred" costs). These are irrelevant to decision-making.
- **Avoidable Cost:** A cost that can be eliminated by choosing one alternative over another (e.g., fixed costs that would disappear if you chose to buy a component instead of making it).

Worked Examples

1. Basic Break-even Calculation

- **Data:** Selling Price (SP) = 600; Variable Cost (VC) = 360; Fixed Cost (FC) = 480,000.
- **Step 1 (Contribution):** $600 - 360 = 240\$$
- **Step 2 (BEP Units):** $\frac{480,000}{240} = 2,000 \text{ units}$
- **Step 3 (BEP Sales):** $2,000 \text{ units} \times 600 = 1,200,000\$$

2. Calculating Units for Target Profit

- **Data:** Target Profit = 270,000; Fixed Cost = 630,000; Contribution = 180 (derived from $500 - 320\$$).
- **Logic:** To find the units required to earn a specific profit, the target profit must be added to the fixed costs to determine the total contribution required.
- **Calculation:** $\frac{630,000 + 270,000}{180} = 5,000 \text{ units}$
- **Total Revenue:** $5,000 \times 500 = 2,500,000\$$

3. Profit Volume (P/V) Ratio & Profit Calculation

- **Data:** Sales = 250,000; VC = 150,000; Fixed Cost = 600,000.

Step 1 (Profit):

1. Contribution = \$250,000 - 150,000 = 100,000\$
2. Profit = \$100,000 - 600,000 = -500,000\$ (Note: Instructor focused on the calculation logic: $\text{Contribution} - \text{Fixed Cost} = \text{Profit}$).

- **Step 2 (P/V Ratio):** $\frac{\text{Contribution}}{\text{Sales}} = \frac{100,000}{250,000} = 0.4$ (or 40%).

4. Finding Fixed Cost from BE Sales and P/V Ratio

- **Data:** BE Sales = 1,800,000; P/V Ratio = 35% (0.35).
- **Calculation:** $1,800,000 \times 0.35 = 630,000$ (Fixed Cost).

5. Margin of Safety (MoS)

- **Data:** Total Sales = 400,000; BE Sales = 250,000.
- **Calculation:** $400,000 - 250,000 = 150,000$ (Margin of Safety).
- **Alternative Formula:** $\frac{\text{Profit}}{\text{P/V Ratio}}$.
- **Note:** The "extra" amount sold beyond break-even is what contributes to profit; therefore, dividing Profit by the P/V ratio isolates the Margin of Safety.

6. Multi-Product (Weighted Average Contribution)

- **Data:** Mix of 3:2. Contribution A = 50; Contribution B = 80. Fixed Cost = 920,000.
- **Step 1 (Bundle Contribution):** $(3 \times 50) + (2 \times 80) = 150 + 160 = 310$.
- **Step 2 (BEP in Bundles):** $\frac{920,000}{310} = 2,968$ bundles.
- **Note:** When calculating for a bundle, the denominator is the sum of the contributions of the specific mix.

7. Impact of Price/VC Changes on BEP

Scenario A (Price Reduction):

- Original: SP = 800, VC = 500, BEP = $120,000 / 300 = 400$ units.
- New: Price reduced by 10% (New SP = 720).
- New Contribution: $720 - 500 = 220$.
- New BEP: $120,000 / 220 \approx 545$ units.
- **Reasoning:** As the denominator (contribution) decreases, the number of units required to break even increases.

Scenario B (VC Increase):

- If VC rises, contribution falls, and BEP units will increase because you must sell more to cover the same fixed costs.

8. Solving for BE Sales and P/V Ratio from Profit and MoS

- **Data:** Sales = 500,000; MoS = 100,000.
- **Step 1 (BE Sales):** $\text{Sales} - \text{MoS} = \text{BE Sales} \rightarrow 500,000 - 100,000 = 400,000$.
- **Step 2 (P/V Ratio):** $\frac{\text{Profit}}{\text{MoS}}$. (Note: Since Profit is only earned on the Margin of Safety, $\text{MoS} = \text{Profit} / \text{P/V Ratio}$).

9. Calculating Profit from MoS and P/V Ratio

- **Logic:** If you multiply the Margin of Safety by the Profit Volume (P/V) ratio, you get the Profit.
- **Calculation:** $\text{MoS} \times \text{P/V Ratio} = \text{Profit}$.

- **Example:** $\$800,000 \times 0.8$ (if P/V is 80%) $\rightarrow$ (The instructor used 800,000 divided by 1,000,000 to demonstrate ratio application).

10. Alternative Calculation Methods (Common denominator)

- **Scenario:** Calculating difference between $\$550,000$ and $\$400,000$ divided by $\$75$.
- **Method A:** $\frac{550,000 - 400,000}{75} = \frac{150,000}{75}$.
- **Method B:** $\frac{550,000}{75} - \frac{400,000}{75}$.
- **Logic:** Both methods are mathematically identical. You can find the difference of the raw figures before dividing, or divide each figure by the denominator first and then find the difference.

11. Finding Fixed Cost from Profit and P/V Ratio

- **Data:** Profit = 6 lakhs; Sales = 360,000; P/V Ratio = 25%.
- **Step 1 (Contribution):** $\text{Sales} \times \text{P/V Ratio} = \text{Contribution}$.
- **Step 2 (Fixed Cost):** $\text{Contribution} - \text{Profit} = \text{Fixed Cost}$.
- **Logic:** Since $\text{Profit} = \text{Contribution} - \text{Fixed Cost}$, it follows that $\text{Fixed Cost} = \text{Contribution} - \text{Profit}$.

12. Calculating Sales for a Target Profit (Using Contribution %)

- **Data:** Contribution Margin = 30%; Fixed Cost = 9 lakh; Profit = 6 lakh.
- **Step 1 (Contribution):** $\text{Profit} + \text{Fixed Cost} = \text{Contribution}$.
- **Step 2 (Sales):** $\frac{\text{Contribution}}{\text{Contribution \%}} = \text{Sales}$.
- **Logic:** Adding profit to fixed cost gives you the total contribution required. Dividing that by the contribution percentage (e.g., 0.30) gives the total sales required.

13. Calculating Pre-tax Profit from After-tax Profit

- **Data:** After-tax profit = 420,000; Tax rate = 30% (0.30).
- **Calculation:** $420,000 \div (1 - 0.30) = 420,000 \div 0.70 = 600,000$ (Pre-tax Profit).
- **Reasoning:** To find the before-tax profit, divide the after-term figure by the percentage of total profit that is retained after tax (1 minus the tax rate).

14. Indifference Point (Comparing Two Machines)

- **Data:** Machine A: Fixed Cost = 500,000; Contribution/unit = 40. Machine B: Fixed Cost = 800,000; Contribution/unit = 60.
- **Logic:** The indifference point is where both machines produce the same profit. To find this, equate the profit equations: $(\text{Contribution} \times Q) - \text{Fixed Cost}$.
- **Calculation:** $40Q - 500,000 = 60Q - 800,000 \rightarrow 20Q = 300,000 \rightarrow Q = 15,000$.
- **Note:** This identifies the volume where both options are equally viable in terms of profit.

15. Multi-Product Bundle Calculation (Detailed)

- **Data:** Sale Product X: Price 200, VC 120; Product Y: Price 300, VC 210. Mix 1:3. Fixed Cost = 720,000.
- **Step 1 (Contribution):** $\$X = 200 - 120 = 80$; $\$Y = 300 - 210 = 90$.
- **Step 2 (Bundle Contribution):** $(1 \times 80) + (3 \times 90) = 350$.
- **Step 3 (Break-even):** $720,000 / 350 = 2,057$ bundles.
- **Reasoning:** When a mix is given, the "unit" of calculation is the bundle. The denominator becomes the total contribution of one "set" of the mix.

16. Impact of Sales Decline

- **Data:** Current Sales = 800,000; Decline = 15%; P/V Ratio = 35%; Fixed Cost = 2,000,000.
- **Step 1 (New Sales):** $800,000 \times 0.85 = 680,000\$$.
- **Step 2 (New Profit):** $\text{\$}\text{New Sales} \times \text{P/V Ratio} = \text{New Contribution}\$$.
- **Step 3 (Final Profit):** $\text{New Contribution} - \text{Fixed Cost} = \text{Profit}\$$.

17. Make or Buy Decisions (Relevant Costs)

- **Scenario A (Sunk Costs):** Purchase Price = 45. Manufacturing cost components $(18+12+8) = 38$. Fixed cost = 10.
- **Logic:** Since the fixed cost of 10 is not stated as "avoidable" (it remains even if you buy), it is not included in the comparison.
- **Decision:** Make $(38 < 45)$.
- **Scenario B (Avoidable Costs):** Purchase Price = 60. Manufacturing costs $(48) + \text{Avoidable Fixed costs } (40) = 88$.
- **Logic:** Because the 40 is specifically avoidable if you choose to make, it is included in the manufacturing total.
- **Decision:** Buy $(88 > 60)$.
- **Critical Logic: Sunk Costs** (costs already incurred) are always ignored. Only **Avoidable Costs** (costs that change based on the decision) should be included in the comparison.

18. Quality Adjustments and Added Costs

- **Data:** Purchase price = 40; Preparation/Quality improvement cost = 30,000; Manufacturing cost = 42.
- **Step 1 (Adjusted Purchase):** $400,000 + 30,000 = 430,000\$$.
- **Step 2 (Make Cost):** $420,000\$$.
- **Decision:** Make.
- **Logic:** Even if the raw purchase price is lower, additional costs incurred to make the product "usable" or "up to quality standards" must be added to the purchase side of the comparison.

19. Opportunity Cost (Labor/Resources)

- **Data:** Manufacturing cost $(35+30+20) = 85$. Purchase Price = 100. Labor can be used elsewhere for 15/unit.
- **Step 1 (Calculate Opportunity Cost):** $85 + 15 = 100\$$.
- **Decision:** Both equal; use non-financial factors.
- **Logic:** If a resource (like labor) is used internally, and it could have generated profit elsewhere, that "lost" potential is an opportunity cost and must be added to the "Make" cost.

20. Subsidy Impact and Multi-Year Contracts

- **Scenario A (Subsidy):** Buy Price = 85; Make Price = 80; Subsidy on purchase = 40.
- **Calculation:** $85 - 40 = 85\$$ vs $80\$$. (Note: In the case of a 100% subsidy on the difference, or specific discount, it reduces the purchase cost).
- **Scenario B (Multi-Year):** Buy = 90 (fixed for 2 years); Make = 70 (variable) + 120 (lease per year).
- **Calculation:** $90 \times 2 = 180\$$ vs $(70 \times 2) + (120 \times 2) = 140 + 240 = 380\$$. (Note: Ensure the timeframe matches for both).
- **Logic:** Subsidies reduce the cost of the selected option. For multi-year contracts, ensure the contract duration is factored in equally for both options.

21. Inflation Adjustment

- **Data:** Current Purchase = 60; Expected Inflation = 10%. Future Purchase = 66.
- **Make Cost:** 58 (fixed).
- **Decision:** Make.
- **Logic:** If a "current" purchase price is given but the decision is for the future, the purchase cost must be adjusted for anticipated inflation before comparing.

22. Step-Fixed Costs

- **Data:** Need 18,000 units. Buy = 52. Make = 40. If production > 15,000, add 1,200,000 supervisor cost.
- **Step 1 (Make):** $\$(18,000 \times 40) + 1,200,000 = 720,000 + 1,200,000\$$.
- **Step 2 (Buy):** $\$18,000 \times 52 = 936,000\$$.
- **Decision:** Make.
- **Logic:** Step-fixed costs are only added to the "Make" cost if the production volume exceeds the specific threshold defined in the problem.

23. Multi-Supplier Comparison

- **Data:** Supplier A = 78; Supplier B = 74 + 50,000 (transport); Make = 72.
- **Comparison:** A = 7,800; B = 7,900; Make = 7,200.
- **Decision:** Make.
- **Logic:** When multiple options exist (e.g., multiple suppliers), calculate the total cost for every option and choose the minimum.

24. Logic of "Switching Sides"

- **Concept:** If a cost is moved from one side of the calculation to the other (e.g., moving a cost from "Buy" to "Make" or vice-versa), the mathematical sign of that item must be flipped.
- **Note:** To avoid confusion, it is often cleaner to calculate the "Total Cost to Make" and "Total Cost to Buy" separately and then compare.

Formulas & Rules

- **Contribution per Unit:** $\text{\$}\{\text{Selling Price}\} - \text{\$}\{\text{Variable Cost}\}$
- **Break-even Point (Units):** $\text{\$}\frac{\{\text{Fixed Cost}\}}{\{\text{Contribution per Unit}\}}$
- **Break-even Point (Sales):** $\text{\$}\{\text{BEP Units}\} \times \text{\$}\{\text{Selling Price}\}$
- **Target Profit (Units):** $\text{\$}\frac{\{\text{Fixed Cost}\} + \{\text{Target Profit}\}}{\{\text{Contribution per Unit}\}}$
- **Profit Volume (P/V) Ratio:** $\text{\$}\frac{\{\text{Contribution}\}}{\{\text{Sales}\}}$ or $\text{\$}\frac{\{\text{Selling Price}\} - \{\text{Variable Cost}\}}{\{\text{Selling Price}\}}$ (expressed as %)
- **Fixed Cost (from BE Sales):** $\text{\$}\{\text{Break-even Sales}\} \times \text{\$}\{\text{P/V Ratio}\}$
- **Margin of Safety:** $\text{\$}\{\text{Total Sales}\} - \text{\$}\{\text{Break-even Sales}\}$
- **Margin of Safety (Alternative):** $\text{\$}\frac{\{\text{Profit}\}}{\{\text{P/V Ratio}\}}$
- **Bundle Contribution:** $\text{\$}\sum (\{\text{Quantity in Mix}\} \times \{\text{Contribution per Unit}\})$
- **Impact Rule:** If the denominator (Contribution) decreases, the BEP units will rise.
- **Operating Leverage:** $\text{\$}\frac{\{\text{Contribution}\}}{\{\text{Profit}\}}$
- **Contribution Ratio:** $\text{\$}1 - \text{\$}\{\text{Variable Cost Ratio}\}$
- **Profit Calculation (from Sales/VC %):** $\text{\$}\{\text{Sales}\} \times (1 - \text{\$}\{\text{Variable Cost Ratio}\}) - \text{\$}\{\text{Fixed Cost}\} = \text{\$}\{\text{Profit}\}$
- **Pre-tax Profit:** $\text{\$}\frac{\{\text{After-tax Profit}\}}{(1 - \text{\$}\{\text{Tax Rate}\})}$
- **Indifference Point:** $\text{\$}\{\text{Contribution}\}_1 - \text{\$}\{\text{Fixed Cost}\}_1 = \text{\$}\{\text{Contribution}\}_2 - \text{\$}\{\text{Fixed Cost}\}_2$
- **Make or Buy Comparison:** $\text{\$}\{\text{Compare (Variable Cost + Avoidable Fixed Cost)}\} \text{\$}\{\text{vs. Purchase Price}\}$
- **Relevant Make Cost:** $\text{\$}\{\text{Direct Materials}\} + \text{\$}\{\text{Direct Labor}\} + \text{\$}\{\text{Variable Overhead}\} + \text{\$}\{\text{Opportunity Cost}\}$
- **Adjusted Buy Cost:** $\text{\$}\{\text{Purchase Price}\} + \text{\$}\{\text{Quality/Fix Costs}\} - \text{\$}\{\text{Subsidies}\}$
- **Opportunity Cost:** The value of the next best alternative (must be added to "Make" cost if the resource is used internally).
- **Subsidies:** Reductions in cost from external sources (subtract from "Buy" cost).
- **Step-Fixed Cost:** A fixed cost that is only incurred when a specific threshold of activity is reached.

Tricky Logic & Traps

- **Pre-tax vs. After-tax:** If a question gives you "after-tax" profit and a tax rate, you **must** calculate the pre-tax profit (divide by $1 - \text{tax rate}$) before you can calculate the required contribution.
 - **Indifference Point vs. Break-even Point:** A common mistake is confusing these. Break-even point is where profit is zero. Indifference point is where the profit of two different options is **equal** to each other.
 - **Sunk Cost Trap:** In "Make or Buy" scenarios, if a cost is described as "already incurred" or "fixed costs that won't change regardless of the decision," it is a **Sunk Cost**. You must exclude these from your calculation.
 - **Avoidable Cost Rule:** In "Make or Buy," only include fixed costs that change based on the decision (Avoidable Costs). If the cost stays the same whether you make or buy, it is irrelevant to the decision.
 - **Denominator Effect:** When price decreases or variable costs increase, the contribution per unit falls (the denominator gets smaller). This makes the quotient larger, meaning you need more units to break even.
 - **Multi-product Bundling:** Do not calculate BEP for each product individually. Use the ratio of the mix to calculate a "bundle contribution." The denominator becomes the total contribution of one "set" of the mix.
- Operating Leverage and Risk:** High operating leverage indicates a high proportion of fixed costs. This creates a "double-edged sword":
- **Upside:** If sales increase, profit increases exponentially.
 - **Downside:** If sales decrease, profit drops much faster because the high fixed costs must still be covered.
- **Contribution Ratio Shortcut:** If you are given the Variable Cost Ratio (e.g., 60%), the Contribution Ratio is automatically $1 - 0.60 = 40\%$. Use this to find the contribution amount from the total sales: $\text{Sales} \times \text{Contribution \%} = \text{Contribution}$.
 - **Opportunity Cost Trap:** If a resource (like labor or space) is used internally, and it *could* have been used to generate profit elsewhere, that "lost" potential is an opportunity cost and **must** be added to the "Make" cost.
 - **Subsidies:** These are discounts or grants that **reduce** the cost of an option (e.g., a subsidy on purchasing reduces the "Buy" cost).
 - **Step-Fixed Costs:** These are costs that only trigger at specific thresholds (e.g., "if production exceeds X units, add supervisor cost"). These are only included if the specific volume is reached.
 - **Non-Financial Factors:** When costs are equal (or nearly equal), use non-financial factors like "Quality" or "Reliability" to make the final decision.
 - **Contract Timing:** When comparing options like 2-year contracts, ensure the time horizon is identical for both options.

Exam Pointers

- **Identifying the correct Profit calculation:** Always start with Contribution (Sales - VC). Profit is then found by subtracting Fixed Costs from that Contribution.
- **Reverse Engineering Fixed Costs:** If you are given Break-even Sales and a P/V Ratio, you can bypass several steps to find Fixed Costs directly by multiplying the two.
- **Two-Product Logic:** When a mix is given (e.g., 3:2), the "unit" of calculation is a "bundle" of 5 units (3 of product A and 2 of product B).
- **Shortcuts for Fixed Costs:** If you know Profit, Sales, and the P/V Ratio, you can find Fixed Cost by calculating: $\text{Sales} \times \text{P/V Ratio} - \text{Profit}$.
- **Shortcuts for Sales:** If you know Profit, Fixed Cost, and the Contribution Margin (%), you can find Sales by: $\frac{\text{Profit} + \text{Fixed Cost}}{\text{Contribution \%}}$.
- **Make/Buy Decision Logic:** Focus on **Relevant Costs**. If a cost is "already incurred," ignore it. Only include "Avoidable Fixed Costs" when comparing the cost of making to the cost of buying.
- **Sales Change Calculations:** When a sales change is given (e.g., "Sales fell by 10%"), calculate the new sales figure first before determining the new profit or other metrics.
- **Quality Adjustments:** If a purchase is cheaper but requires extra "fixing" or "quality" costs, add those to the Purchase price.
- **Multi-Option Selection:** When multiple vendors/options exist, calculate the total cost for each and choose the minimum.

- **Inflation Logic:** If a "current" purchase price is given but the decision is for the future, adjust the purchase cost for the predicted inflation before comparing.
- **Exam Resource Materials:** The provided exam PPT contains the necessary formulas (Break-even, PV Ratio, etc.).
- **Allowed Resources:** You are permitted to bring one textbook and printed copies of the exam PPT.
- **Examination Scope:** The exam is based specifically on the provided PPTs.

Sources

- PV ratio practice question.docx
- Make or Buy and Special Order.docx